

# ANAKIN DEY

331-250-1378 | [anakind2@illinois.edu](mailto:anakind2@illinois.edu) | [anakin-dey.com](http://anakin-dey.com) | [github.com/spamakin](https://github.com/spamakin) | [linkedin.com/in/anakin-dey](https://linkedin.com/in/anakin-dey)

## EDUCATION

### University of Illinois Urbana-Champaign

Bachelors of Science in Mathematics, Minor in Computer Science

May 2024

GPA: 3.91/4.00

## EXPERIENCE

### Software Engineering Intern: CME Group

Jun 2022 – Aug 2022

- Refactored large code bases in order to support multiple file formats and platforms for tracing and logging of messages, resulting in tracing having 20% less impact on the performance of applications
- Used Jenkins, XL Release, and in-house tools to benchmark the performance impact from using the new formats
- Explored the use of open-source dashboards and parsers to better process traces and give new insights to data

### Learning, Decision, Control, Autonomy Lab: UIUC

Aug 2021 – Present

- Researching algorithms and heuristics for multi-target and multi-agent planning and movement under multiple constraints as well as deceptive and adversarial movement over graphs
- Combining concepts from heuristic approximation algorithms, Markov decision processes, probability theory, geometry, and graph theory to design new and efficient heuristics to use with various algorithms
- Benchmarking algorithms in Python and using Matplotlib, NetworkX, and OSMnx to create diagrams and charts
- Maintaining high code quality with various tools and workflows such as Black, pytest, Flake8, and mypy

### Software Engineering Intern: SiteIQ

May 2021 – Aug 2021

- Improved the accuracy of status reports from multiple fueling stations using Python and JavaScript by streaming and parsing data sources such as two-wire serial and ZMODEM files
- Identified error codes and data that were not described in given documentation but present in 50% of devices and logged them for further analysis by the team
- Presented findings to managers and coworkers with multiple levels of technical understanding

### Course Assistant: UIUC

Jan 2021 – Present

- Course Assistant for Introduction to Computer Science (Jan 2021 - May 2021), Discrete Structures (Aug 2021 – Present), Introduction to Algorithms (Jan 2022 – Dec 2022) and Models of Computation (Jan 2023 - Present)
- Teaching theoretical computer science concepts such as state machines, divide-and-conquer, dynamic programming, and graph algorithms by presenting topics in a clear and concise manner
- Holding office hours and helping students in class forums of 400+ students with both concepts and homework

## PUBLICATIONS

### Post-Disaster Repair Crew Assignment Optimization Using Minimum Latency

Anakin Dey, Melkior Ornik ([IEEE IS2C 2022](#))

- Designed a new multi-agent heuristic algorithm to solve a weighted path planning problem in the context of outage restoration after natural disasters which was up to 50% better than existing heuristics
- Implemented and tested a graph class and algorithms in Python using pytest and Coverage.py for code validation
- Paper on [arXiv](#) and source code on [GitHub](#)

## EXTRACURRICULARS

### Founder & President: SIGma | Special Interest Group for Math and Algorithms | [csttheory.org](http://csttheory.org)

- Founded club to teach theoretical computer science and mathematics to students of various backgrounds
- Creating weekly meetings covering ideas such as efficient [binary generation](#), [decidability](#) and Turing Machines, [state machines](#), and other [mathematical](#) concepts
- Created meetings with the aim of making these concepts more intuitive and enjoyable to learn and with the end goal of fostering a genuine interest in the subject matter

### Admin: SIGPwny | Special Interest Group for Security | [sigpwny.com](http://sigpwny.com)

- Hosted [2 meetings](#) on cryptographical concepts including block ciphers, RSA, Diffie-Hellman, and elliptic-curve cryptography, complete with multiple challenges, to over 70 people per meeting
- Created cryptography challenges and reverse engineering for [UIUCTF 2021](#) / [2022](#) for over 1400 participants and [Fall CTF 2021](#) / [2022](#) for over 300 participants

## TECHNICAL SKILLS

**Languages:** Python, Java, C++, JavaScript, TypeScript, Agda, C

**Tools / Frameworks:** Bash, Git, JUnit, mypy, pytest, Matplotlib, NetworkX, Numpy, GDB, Catch2, Node.js, Pandas, Jira, Grafana